



Lesson 5: While Loops and For Loops



Welcome Back!

Remember Scratch's "repeat" block, that let your sprite do something over and over without stacking the same block 10 times? Python has that superpower too — it's called a **loop**!



Think Like a Coder

A loop is like telling a robot "do this again and again" — either a set number of times (`for`), or until something changes (`while`).



Step 1: Your First For Loop

Type and run:

```
for i in range(5):  
    print("Blast off in", i)
```

Look closely at the output — it counts 0, 1, 2, 3, 4. That's still 5 numbers total — Python just starts counting at 0!



Robot Tip

`range(5)` means "5 numbers, starting at 0" — so it stops at 4, not 5! If you want to count 1 through 5, use `range(1, 6)` instead.



Step 2: A Countdown

```
for i in range(5, 0, -1):  
    print(i)  
print("Liftoff!")
```

This counts DOWN from 5 to 1, then prints "Liftoff!"

Step 3: Your First While Loop

```
count = 1  
while count <= 3:  
    print(f"Lap {count}")  
    count = count + 1
```

A `while` loop keeps going as long as its condition is true. Here, it repeats while `count` is 3 or less.

Debug Detective

This code never stops running! Can you figure out why?

```
count = 1  
while count <= 3:  
    print(f"Lap {count}")
```

Hint: does anything inside the loop ever change?

The Fix:

```
count = 1  
while count <= 3:  
    print(f"Lap {count}")  
    count = count + 1
```

The original code never changed `count`, so `count <= 3` was ALWAYS true — an infinite loop! We must update the variable being checked, or the loop never stops. (If this ever happens by accident, press the red Stop button in Thonny.)

Mini Challenge

Change the `for` loop to count from 1 to 5 instead of 0 to 4:

```
for i in range(1, 6):  
    print("Blast off in", i)
```

Bonus Challenge

Combine a `while` loop with `input()` to keep going until the player types "quit":

```
answer = ""  
while answer != "quit":  
    answer = input("Type a word (or 'quit' to stop): ")  
    print(f"You typed: {answer}")  
print("Goodbye!")
```

Make It Yours

Build "The Rocket Countdown" — check [Activities.md](#) for full steps!

Victory Moment

You just taught your program to repeat itself without writing the same line over and over — a huge coding superpower!

Coming Up Next

In Lesson 6, we'll learn about **lists** — a way to store lots of information (like a whole crew of astronauts) in one single variable!